

Response to Prescreening Comments

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Summary of the revision

The prescreening comments identified that the problem formulation, technical novelty, and primary scientific contribution of the submission were not sufficiently clear. We have revised the manuscript accordingly. The work is now positioned as a general evaluation contract, FAIR-X, with FAIR-BETH as its executable instantiation on the BETH dataset. The same contract is also demonstrated on MalBehavD-V1. The manuscript no longer presents the work as a new malware detector or a deployment-ready product, but as a reproducible methodology for deciding when behavioral scores support dependable alerting decisions under distribution shift.

The full reproducibility package, including scripts, generated CSV/JSON artifacts, figures, and LaTeX sources, is available at:

<https://github.com/EkodeckStephane/fair-beth-malicious-process-detection>

Point-by-point response to the Associate Editor’s comments

Comment 1: *“The problem formulation, technical novelty, and primary scientific contributions are not sufficiently clear.”*

Response. We have rewritten the introduction, abstract, and conclusion to state the problem explicitly. The central problem is *threshold transfer*: how to convert behavioral security scores into deployable decisions when validation data are scarce and the deployment distribution differs from the validation distribution. The primary scientific contribution is the FAIR-X evaluation contract, which fixes data access, label exclusion, feature-vocabulary isolation, common threshold policies, strong baseline parity, uncertainty reporting, and failure-boundary disclosure before any final-test evaluation. The contract is instantiated as FAIR-BETH on BETH, but the invariants are defined independently of any dataset. Three quantitative diagnostics—tabular-dominance gap, score-compression gap, and threshold sensitivity—are introduced to make fusion and threshold-transfer claims falsifiable.

Revised sections: Introduction (Section 1), Problem Formulation (Section 2), FAIR-X Contract (Section 3), and Contribution Summary (Section 6).

Comment 2: *“It remains unclear whether the work introduces a fundamentally new detection framework, a principled theoretical analysis, or a generalizable security evaluation methodology beyond the specific datasets and experimental settings considered.”*

Response. The revised manuscript makes explicit that it does *not* introduce a new detection framework and does not claim a principled theoretical analysis in the formal sense. Its contribution is a *generalizable security evaluation methodology*. FAIR-X is presented first as a dataset-independent contract; only afterwards is it instantiated on BETH. The same contract is then executed on MalBehavD-V1, using an entirely different feature schema (API-call sequences), to demonstrate portability. The generalizability comes from the invariants, not from the specific models or datasets.

Revised sections: Section 3 (FAIR-X contract), Section 7 (External Validation on MalBehavD-V1), and Section 8 (Operational Implications and Limitations).

Comment 3: *“The relationship to existing malware/process behavior evaluation methodologies also needs to be better established.”*

Response. The related-work section now distinguishes FAIR-X from existing security-ML evaluation guidance. Prior work identifies common pitfalls (leakage, optimistic thresholding, weak baselines, distribution shift) and proposes best practices. FAIR-X goes further by *operationalizing* several of these concerns into an executable audit protocol: it specifies exactly which records each stage may access, which threshold policies apply to every model, which non-hybrid baselines must be matched, and which diagnostics must be reported. The comparison with Highnam et al. (BETH) and with API-call sequence models on MalBehavD-V1 is now framed as an application of this protocol, not as a state-of-the-art accuracy claim.

Revised sections: Section 2 (Related Work) and Section 9.1 (Comparison with Original BETH Work).

Comment 4: *“The broader implications for real-world malicious-process detection and the general applicability of the proposed protocol require further justification.”*

Response. A new operational-implications section explains the real-world consequences of the findings. It argues that ranking metrics (AUC/AP) are insufficient for dependable endpoint decisions; that validation-derived thresholds can fail under deployment shift; that environment-specific calibration is necessary; and that operational response claims require additional evidence on prefix latency, cost asymmetry, response reversibility, and analyst workload. The section also states the model-card limitations explicitly: the evaluated detector should produce alerts and telemetry-enrichment requests, not autonomous process termination, unless paired with independent high-confidence signals.

Revised sections: Section 8 (Operational Implications and Limitations) and Section 10 (Model Card).

Comment 5: *“Therefore, the technical contribution and impact of the work are difficult to fully assess.”*

Response. We have sharpened the technical contribution in three ways. First, the FAIR-X contract is now stated as a list of explicit invariants and an algorithm

(Algorithm 1). Second, the three diagnostic gaps are defined mathematically and reported numerically for every supervised model. Third, the revised manuscript adds extensive audits that make the contribution assessable: repeated cross-fitted threshold selection over all 29 BETH development positives, target-copy-free GRU/LSTM sequence ablations, additional tabular baselines, calibration audit, MalBehavD-V1 external protocol validation, prefix evaluation, cost sensitivity, feature-space stress tests, and host/temporal split robustness. The claim boundaries are also made explicit: the work does not claim ransomware-family attribution, calibrated posterior risk, direct BETH-to-MalBehavD model transfer, or autonomous containment readiness.

Revised sections: Section 3 (FAIR-X Contract and Diagnostics), Section 5 (Experiments), Section 7 (External Validation), Section 9 (Failure Mode Analysis), and Section 10 (Model Card).

Key empirical corrections

All experiments were rerun with aligned methodologies. Under the common validation-FPR policy on the official BETH process-level test split the corrected results are:

- TabRF: F1 0.7210
- RF-500 (now using 500 trees): F1 0.7043
- MetaRF: F1 0.6756

TabRF retains the highest F1 point estimate, but its confidence interval overlaps RF-500’s interval, and MetaRF’s interval overlaps TabRF’s. The diagnostic conclusion is therefore bounded: the experiment provides no evidence that MetaRF improves over the simpler tabular alternatives.

The repeated cross-fitted threshold audit fixes 50 thresholds from all 29 BETH development positives before official-test access. Its locked median threshold yields F1 0.6964 but expands the 5% development-FPR target to 32.47% test FPR. On MalBehavD-V1, the same FAIR-X contract gives a repeated-split median F1 of 0.9533.

All numerical values in the revised manuscript and supplement are synchronized with the CSV/JSON artifacts in the submitted reproducibility package.

List of major changes

- Manuscript retitled to emphasize the general FAIR-X contract.
- Introduction and conclusion rewritten around threshold transfer and claim boundaries.
- FAIR-X contract formalized as invariants and Algorithm 1.
- Three quantitative diagnostics defined and reported.
- Related work expanded to distinguish FAIR-X from existing evaluation guidance.
- Operational implications and model-card limitations added.
- RF-500 baseline corrected to 500 trees; all tables, figures, and CSV artifacts regenerated.
- Threshold policy unified and documented across all scripts.
- Extensive audits added or clarified: repeated thresholds, sequence ablation, calibration, MalBehavD-V1, prefix, cost, stress, host/temporal splits.
- Reproducibility package updated with scripts, environment details, and generated artifacts.